

Instructions for the System code practical session:

To perform the calculations you will need to connect to a server from your Operating System (OS). All steps on the server will be carried out through a Terminal in the FTP tool (e.g. WinSCP). So you will need to manage the basic UNIX terminal commands.

In order to be able to perform all the tasks you need to install the following programs:

- First you need to connect to UPCLink (this is a VPN that allows to connect to UPC services): <https://serveistic.upc.edu/ca/upclink/documentacio>
- A good text editor like: **notepad++**, vim, sublime, VScode. If you don't have any, I recommend to install notepad++:
 - <https://notepad-plus-plus.org/> it is free opensource and has what you need to edit text
- An FTP program to transfer files from/to the server:
 - WinSCP is good <https://winscp.net/eng/download.php>
 - Xftp+Xshell is much better and I believe it has a free license for Universities <https://www.netsarang.com/en/free-for-home-school/>
- An SSH program to connect to the server from your OS:
 - In windows 11 you can do it directly from Powershell prompt
 - Xshell
 - You can use WinSCP to show the directories in the server and to execute commands in the server.
 - You can install UNIX Terminal emulators, e.g. **putty**, **Cygwin** or use the **Ubuntu Terminal** from Windows 10. Note that those emulators allow you to execute commands on the server, and file transfer from/to the server is done through FTP.
 - If your OS is Linux or MAC, you can install an SSH interpreter and use your OS directly. Another FTP tool for Linux is FileZilla.

Connecting to the server:

Herein, commands will be highlighted in Yellow, and variables in commands that need to be changed will be capitalized.

- Connect to UPCLink as per instructions on the UPC website (the first link in this document)
- If you are using WinSCP, the program will ask you for the IP address, Username and Password of the server when you connect.
- **IMPORTANT** make sure that the VPN (UPCLink) is active.
- Server IP: 10.5.18.62
- Port:3322
- Username: You tell me in class....
- You will be prompted to enter the password
 - **THISgreat2024**

- You will be prompted to change the password. Change it (at least 8 characters with upper, lower case and symbol/number)
- Connect again
- If you are NOT using WinSCP to execute commands in the server, then open your SSH method (e.g. Linux Terminal if your OS is Linux) and execute the following command:
 - `ssh 10.5.18.62 -p 3322 -l USERNAME`

where USERNAME is that assigned to your group. The terminal will automatically ask you for password after execution

Running the case:

- Once you have finished your input, move it to the server using the FTP tool (WinSCP)
- In the folder where you have the input, run the command (you must be familiar now with either a built-in Terminal, UNIX Terminal emulator or Terminal feature in WinSCP):
 - `relap1.sh YOUR_INPUT run`
- Note that YOUR_INPUT is the name of your input file without the extension (YOUR_INPUT.i)

Plotting and extracting data in a text file:

Option A (no longer available): Plot with APTPLOT:

To plot your results in your Operating System (OS), e.g. Windows 10, using APTPLOT, move the file with (*.r) extension from the server directory to a directory in your OS using the FTP tool (WinSCP). In the interface of APTPLOT, go the menus and select the following:

- File >> read >> navigate to the directory where you have transferred the YOUR_INPUT.r file from the server >> select the file.

Option B, extract data:

- Prepare your list of variables to extract: STRIP_FILE.i and place it at the folder where you have the file with all the results (YOUR_INPUT.r)
- Note that STRIP_FILE.i is not the same file as YOUR_INPUT.i
- Run:
 - `extract.sh STRIP_FILE YOUR_INPUT`
- Now, you have your data in a text file: strip_file_your_input.dat
- Using the FTP tool, move the *.dat file to your OS from the server.
- You can also run python commands on the server to get directly the plotting.

Option C, use our `relap_py` python package (part of it need to be executed on the server):

- Go to <https://pypi.org/project/relap-py/> to see a full tutorial
- For installation do the following:
 - `pip install relap_py`
 - `pip install pandas`
 - `export RELAP_EXE_PATH= /usr/local/bin/relap5-33iy-linux-ifc-opt.x`
- I suggest to generate a python file for the data extraction to be done in the server and another one for the plotting locally in your computer.